

AWGN_generator

```
1  import numpy as np
2  import Converter_dB as conv_dB
3
4
5  def add_AWGN_into_sequence(signal_seq, snr_db):
6
7      power_signal = np.mean(np.power(np.abs(signal_seq), 2))
8      snr_linear = conv_dB.convert_SNR_dB_to_SNR(snr_db)
9      power_noise = power_signal / snr_linear
10
11     if np.iscomplexobj(signal_seq):
12         noise = np.sqrt(power_noise/2) * (np.random.randn(*signal_seq.shape)
+ 1j*np.random.randn(*signal_seq.shape))
13     else:
14         noise = np.sqrt(power_noise) * np.random.randn(*signal_seq.shape)
15
16     return signal_seq + noise
17
```